

Maxwell A. Fine

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Education

PhD in Physics & Astrophysics — McGill University (in progress): Advisors Prof. Vicky Kaspi, Prof. Jason Hessels
MSc in Astronomy & Astrophysics — University of Amsterdam: Advisors Prof. Ziggy Pleunis, Prof. Jason Hessels
HBSc in Physics & Astrophysics — University of Toronto

Selected Publications

O.S. Ould-Boukattine, A.J. Cooper, J.W.T. Hessels, D.M. Hewitt, S.K. Ocker, A. Moroianu, K. Nimmo, M.P. Snelders, I. Cognard, T.J. Dijkema, **M. Fine**, et al., (2025). “A HyperFlash and ECLAT view of the local environment and energetics of the repeating FRB 20240619D.” *Submitted to the Monthly Notices of the Royal Astronomical Society*. [arXiv:2509.16374](https://arxiv.org/abs/2509.16374).

Kaitlyn Shin, Alice Curtin, **Maxwell Fine**, Ayush Pandhi et al., (2025). “The CHIME/FRB Discovery of the Extremely Active Fast Radio Burst Source FRB 20240114A.” *Submitted to the American Astronomical Society*. [arXiv:2505.13297](https://arxiv.org/abs/2505.13297).

Maxwell A. Fine, Cameron L. Van Eck, & Luke Pratley. (2023). “Correcting Bandwidth Depolarization by Extreme Faraday Rotation.” *Monthly Notices of the Royal Astronomical Society*. [arXiv:2302.03134](https://arxiv.org/abs/2302.03134).

Selected Research Projects

Deciphering the Local Environments of Repeating Fast Radio Bursts (FRBs) Using Scintillation Sept 2024 – July 2025
University of Amsterdam

Master’s Thesis; Advisors: Prof. Ziggy Pleunis & Prof. Jason Hessels

- Using scintillation to study the local environments of repeating FRBs detected by CHIME, focusing on their emission size regions, and constraining the locations of interstellar, intergalactic scintillation and scattering screens.
- Paper in prep.
- **Skills:** Python, Bash, Linux, Git, time-series analysis, statistical modeling

Developed a Real-Time FRB Detection Pipeline — Dwingeloo Radio Telescope June 2024 – August 2024
Astron Summer Graduate Student Research Fellow; Advisors: Dr. Tammo Jan Dijkema Astron & JIVE & Prof. Jason Hessels

- Operated the 25m [Dwingeloo Radio Telescope \(DRT\)](#) and developed a real-time detection pipeline for repeating FRBs in Python; see the [project’s GitHub repository](#). Pipeline processes data at ~ 1 Gb/s.
- Detected a burst from [FRB20240619D](#), showcasing the telescope’s enduring scientific capabilities. The telescope is now used in FRB follow-up campaigns, and the results will be published in an upcoming paper.
- **Skills:** Python, Linux, Git, Bash, data processing, software development, signal processing, Convolutional Neural Network (CNN), real-time data processing, docker

Multi-wavelength Search for Fast Radio Bursts with Swift/BAT May 2022 – April 2023
SURP Summer Undergraduate Research Fellow; Supervisors: Dr. Ziggy Pleunis, Dr. Paul Scholz University of Toronto

- Conducted a multi-messenger search for X-ray and gamma-ray counterparts to CHIME/FRBs using Swift/BAT.
- Developed a pipeline in Python using HEASoft (written in Bash), and XSPEC for fluence modeling.
- **Skills:** Python, Bash, Git, Linux, XSPEC, HEASoft, time-series analysis

Rotation Measure Synthesis and Error Analysis for Radio Polarization Surveys May 2020 – August 2021
SURP Summer Undergraduate Research Fellow (2x); Advisor: Dr. Cameron L. Van Eck University of Toronto

- Created a novel RM synthesis algorithm for sources with extreme bandwidth depolarization and contributed to the open-source Python package [RM-Tools](#).
- Co-authored a first-author paper published in *Monthly Notices of the Royal Astronomical Society* based on this work.

- Later tested and improved an error analysis pipeline for the [POSSUM](#) survey, identifying and implementing a fix for underestimated polarization uncertainties.
- **Skills:** Python, Git, RM synthesis, error analysis, simulation, algorithm development, big data

Awards & Fellowships

Mcgill Wolfe Fellowship in Scientific & Technological Literacy (\$6,000)	2025
ASTRON Summer Graduate Research Fellowship (€2,500 + Housing)	2024
UofT SURP Research Fellowship (\$28,595 total)	2020, 2021, 2022
John Pounder Prize in Astronomy (3rd Year) (\$200)	2021
NSERC Undergraduate Student Research Award (\$6,000)	2021
Student Excellence and Leadership Award (\$250)	2020
John Pounder Prize in Astronomy (2nd Year) (\$300)	2019

Supervision & Teaching

Research Supervisor , CHIME/FRB Collaboration <i>Supervising: Max Levine</i>	Summer 2026
Teaching Assistant (TA), McGill PHYS241: Signal Processing PHYS101: Introduction to Physics I for the Life Sciences	Winter 2026 Fall 2025
Teaching Assistant (TA), University of Toronto PHYA11: Introduction to Physics I for the Life Sciences PHYA22: Introduction to Physics II for the Life Sciences PHYA10: Introduction to Physics I for the Physical Sciences	Fall 2020 Winter 2021 Fall 2021, Fall 2020
Facilitated Study Group (FSG) Leader, University of Toronto PHYA10: Introduction to Physics I for the Physical Sciences PHYA21: Introduction to Physics II for the Physical Sciences	Fall 2020, Fall 2019 Winter 2020

Conferences and Workshops Attended

CASCA 2026 (Montréal)	June 2026
Radio astronomy workshop (McGill, Montréal)	May 2026
Scintillation 2025 (McGill, Montréal)	Oct 2025
FRB 2025 (McGill, Montréal)	Jul 2025
Scintillometry 2024 (Orlando, University of Central Florida)	Oct 2024
Fast Radio Burst follow-up workshop (Toronto, University of Toronto)	Apr 2023

Service & Outreach

Maintainer, CHIME/FRB Heartbeat Service	August 2025–Present
Maintainer, CHIME/FRB Exposure Pipeline	Jun 2025–Present
Judge - McGill Physics Hack-a-thon, Weismann Science Fair	Winter 2026
Local Organizing Committee, Scintillometry 2025 (McGill University)	2025
Volunteer, 25m Dwingeloo Radio Telescope (DRT)	Summer 2024
Executive & Co-founder, University of Toronto Amateur Astronomy Society (UTAAS)	2022–2023
Director for Physics, UofT Scarborough Campus Student Union	2021–2022
Director for Physics & Astrophysics, UofT Environmental & Physical Sciences Student Association	2018–2020